# Game Theory
# Lecture 3[1]

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# Outline


1. Existence of Nash equilibria
2. Maths: Fixed point theorems
3. Nash theorem

# Existence of Nash equilibrium I

- not all games have a Nash equilibrium

## Example

- 2 players ($N = \{P1, P2\}$)
- each player says a number ($A_i = \mathbb{R}$)
- player that says the higher number wins
  (e.g. winner has payoff 1 while loser has payoff 0, if both say the same number each has payoff 1/2)

# Existence of Nash equilibrium II

Why are we interested in existence?

- worthwhile to search for an equilibrium
- steady state interpretation of equilibrium, existence of NE says that the process might become stable
- sometimes possible to do comparative statics without computing the NE (only sensible if NE exists)

# Fixed point theory I

**Definition (fixed point)**

Let $f$ be a real function, i.e. $f : A \to \mathbb{R}$ where $A \subseteq \mathbb{R}$. A point $a \in A$ is a *fixed point* of $f$ if $f(a) = a$.

- Does $f(x) = 1$ have a fixed point? What about $f(x) = x + 1$? What about $f(x) = x^2$?
- fixed point theory gives general conditions under which functions have fixed points

# Fixed point theory II

**Theorem (Brouwer's fixed point theorem, 1-dimensional)**

*Let $f : [0, 1] \to [0, 1]$ be a continuous function. Then $f$ has a fixed point.*

# Fixed point theory III

## Example (2 player Cournot)

- two firms each choose a quantity $q_i$
- both firms have costs $c(q_i) = cq_i$ for some $c > 0$
- inverse demand is $P(q_1 + q_2)$ where we assume that $P$ is two times continuously differentiable with $P' < 0$ and $P'' \leq 0$
- assume that $P(1) < c$
  $\Rightarrow$ a firm will never offer a quantity greater than 1
- firm 1 chooses a quantity from $[0, 1]$ to maximize profits

$$\max_{q_1} (P(q_1 + q_2) - c) q_1$$

we get the first order condition

$$P'(q_1 + q_2) q_1 + P(q_1 + q_2) - c = 0$$

# Fixed point theory IV

## Example (2 player Cournot continued)

- the second order condition holds by assumption

$$P''(q_1 + q_2)q_1 + 2P'(q_1 + q_2) < 0$$

- the first order condition defines a best response function $q_1(q_2)$
- the best response function is continuous because $P$ and $P'$ are continuous by assumption
- Brouwer: best response function has a fixed point!
- game is symmetric $\Rightarrow$ fixed point is an equilibrium $\Rightarrow$ equilibrium exists
- We showed this without being able to actually calculate the equilibrium!!!

# Fixed point theory V

**Theorem (Brouwer fixed point theorem)**

*Let $S$ be a convex and compact set in $\mathbb{R}^n$ and let $f : S \to S$ be a continuous function. Then there exists an $s^* \in S$ such that $f(s^*) = s^*$.*

# Nash theorem

**Theorem (Nash theorem)**

*A strategic game $\langle N, (A_i), (u_i) \rangle$ with*

- *a finite number of players*
- *a finite number of actions for each player*

*has a Nash equilibrium in mixed strategies.*

- proof uses Brouwer's fixed point theorem
- to simplify notation, we prove the theorem for $2 \times 2$ games

# Proof of Nash theorem for $2 \times 2$ games I

- $2 \times 2$ game:

| | L | R |
|---|---|---|
| U | a,b | c,d |
| D | e,f | g,h |

- mixed strategy of P1: probability $\alpha \in [0, 1]$ of playing U
- mixed strategy of P2: probability $\beta \in [0, 1]$ of playing L
- idea of proof:
  - define a function $f(\alpha, \beta)$ such that
    - $f$ is continuous
    - if $f(\alpha^*, \beta^*) = (\alpha^*, \beta^*)$, then $(\alpha^*, \beta^*)$ is a Nash equilibrium of the game
  - use Brouwer's theorem to establish that $f$ has a fixed point

# Proof of Nash theorem for $2 \times 2$ games II

- $u_1(U, \beta) = \beta a + (1 - \beta)c$ is the expected utility of P1 when playing U and P2 uses the mixed strategy $\beta$
- $u_1(U, \beta)$ is linear and therefore continuous in $\beta$
- define

$$g(\alpha, \beta) = max\left\{0, \frac{\alpha + u_1(U, \beta) - u_1(D, \beta)}{1 + |u_1(U, \beta) - u_1(D, \beta)|}\right\}$$

  - $g(\alpha, \beta)$ is higher than $\alpha$ if U is the best response to $\beta$ and lower than $\alpha$ if D is best response
  - $g$ is continuous because $u_1$ is continuous in $\beta$
- define

$$h(\alpha, \beta) = max\left\{0, \frac{\beta + u_2(L, \alpha) - u_2(R, \alpha)}{1 + |u_2(L, \alpha) - u_2(R, \alpha)|}\right\}$$

# Proof of Nash theorem for $2 \times 2$ games III

- $h(\alpha, \beta)$ is higher than $\beta$ if L is best response to $\alpha$ and lower than $\beta$ if R is best response
- $h$ is continuous because $u_2$ is continuous in $\alpha$

- let $f$ be

$$f(\alpha, \beta) = (g(\alpha, \beta), h(\alpha, \beta))$$

- if $f(\alpha^*, \beta^*) = (\alpha^*, \beta^*)$ then
  - $g(\alpha^*, \beta^*) = \alpha^* \Rightarrow \alpha^*$ is best response to $\beta^*$
  - $h(\alpha^*, \beta^*) = \beta^* \Rightarrow \beta^*$ is best response to $\alpha^*$
  - $(\alpha^*, \beta^*)$ is Nash equilibrium

  every fixed point of $f$ is Nash equilibrium
- $f$ is continuous because $g$ and $h$ are continuous
- Brouwer: $f$ has a fixed point

# Existence revisited I

## Example (revisited)

- 2 players ($N = \{P1, P2\}$)
- each player says a number ($A_i = \mathbb{R}$)
- player that says the higher number wins
  (e.g. winner has payoff 1 while loser has payoff 0, if both say the same number each has payoff 1/2)

- why does the Nash theorem not apply?

# Existence revisited II

Note that the propositions give sufficient conditions for NE, i.e. NE can exist even if the conditions are not satisfied:

**Example (war of attrition 1, OR ex. 18.4)**

- 2 players dispute an object
- time is a continuous variable starting at 0 and running indefinitely
- actions: chose a time $a_i$ when to concede object
- player $i$ attaches value $v_i$ to object and loses one payoff unit for every time unit (until one of the players concedes)

action space is not finite

# Review Questions

- Why are we interested in the question "Does a Nash equilibrium exist?"?
- What is a "fixed point"?
- Explain Brouwers fixed point theorem (1-dimensional case)!
- What does Nash's theorem state?

reading: MSZ 5.3, (supplementary *reading: OR 2.4)

# Exercises I

1. We look at a 2 player game in which $A_1 = A_2 = [0, 1]$. Assume that P1 has a best response function that is strictly increasing and continuous. For all subquestions draw the best response functions in an $a_1$, $a_2$ diagram.
   1. First, assume that P2 also has a best response function that is continuous. Does this guarantee the existence of a Nash equilibrium?
   2. Second, assume that P2 also has a best response function that is increasing (though not necessarily continuous). Does this guarantee the existence of a Nash equilibrium?
   3. Third, assume that P2 has a best response function that is discontinuous at one point only. Does this guarantee the existence of a Nash equilibrium?
2. *Use the Brouwer fixed point theorem to prove that a pure strategy Nash equilibrium exists in the following strategic game:
   - $N = \{P1, P2\}$

# Exercises II

- $A_1 = A_2 = [0, 1]$
- $u_1(a_1, a_2)$ is differentiable and strictly concave in $a_1$. The partial derivative $\frac{\partial u_1}{\partial a_1}(a_1, a_2)$ is continuous in both variables.
- $u_2(a_1, a_2)$ is differentiable and strictly concave in $a_2$. The partial derivative $\frac{\partial u_2}{\partial a_2}(a_1, a_2)$ is continuous in both variables.

*Hint: This exercise is an extension of the Cournot example we did in the lecture. The argument is similar.*